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Where Minds Pollinate

Physics
Class 11
FBISE

Chapter 14
Handwritten Notes
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PHYSICS

CHAPTER : 14

PARTICLE PHYSICS

Elementary particles:

A particle that is not made up of other particles is called elementary particles.

Elementary particles are divided into two groups.

1. **FERMIONS:**

Those fundamental particle with spin halfinteger $\hbar$.

- Fermions follow Pauli exclusion principle.
- Fermions follow fermi - Dirac statistics
- They are matter particles.

2. **BOSONS:**

Those particle with spin integer of $\hbar$.

- Boson do not follow Pauli Exclusion principle.
- Bosons follow Bose. Einstein statistics.
- They are force carriers. (they contain matter particles)
- e.g Graviton and photon are mass less

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Boson.

QUARK THEORY:

In 1964 Murry Gell-Mann and George Zweig proposed the quark theory which hypothesized that all hadrons are composed of a combination of elementary particles called quarks.

=> These elementary particles are arranged in a table known as standard model of elementary particles.

=> Quarks are the elementary particles and basic building block of Menson and Baryon.

=> Every quark ~~has~~ has its anti-quark with same properties but opposite charge.

=> Quarks can be divided into six flavours

1-up 2-down 3-charm 4-strange 5-Top 6-bottom

All quarks have half spin (1/2)

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1st Generation

Up-Quarks : (U)

- These are the most stable and lighest quark.
- charge on up quark is $+\frac{2}{3}e$
- mass range are from $1.7 MeV/c^2 - 3.1 MeV/c^2$
- Anti-up quark is denoted by $\bar{U}$ having charge $-\frac{2}{3}e$

Down Quark: (d)

- These are heavier than up quark but lighter than other, so they are also highly stable.
- charge on down quark is $-\frac{1}{3}e$
- mass range are $4.1 MeV/c^2$ to $5.7 MeV/c^2$.
- Anti-down ~~corle~~ quark is denoted by $\bar{d}$ having charge $+1/3e$.

2nd Generation

Charm Quarks : (C)

- It is hundred times more massive than up and down quarks.
- The charge on charm quark is $+\frac{2}{3}e$
- mass of charm quark is $1280 MeV/c^2$
- Anti charm quark is denoted by $\bar{C}$ having

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2nd generation

charge $-\frac{2}{3}e$.

Strange- Quark : (s)

- The charge on strange cork is $-\frac{1}{3}e$.
- mass of strange quark is 96 MeV/c^2
- Anti- strange quark is denoted by $\bar{s}$ having charge $+\frac{1}{3}e$.

3rd generation

Top - Quark (t):

- These are heavier quark among all the quarks and unstable particle.
- The charge on top quark is $+2/3e$
- mass of top quark is ~~173.1~~ 173.1 GeV/c^2
- Anti-Top quark is denoted by $\bar{t}$ having $-2/3e$.

Bottom - Quark (b):

- Bottom quark decays mostly into a charm quark but rarely into an up quark.
- The charge on bottom quark is $-1/3e$
- The mass of bottom quark is 4.18 GeV/c^2
- Anti- bottom quark is denoted by $\bar{b}$ having charge $+1/3e$.

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COLOR PROPERTY OF QUARKS AND GLUONS

Color is the quantum property of quark but not actual visual color in the theory of "Quantum chromo dynamics"

-> Quarks can be found in three color change.

- Red
- Green
- blue

Their anti-color changes are

- Anti-red
- Anti-Green
- Anti-blue

-> free quarks have never been observed directly. They are always found in groups called Baryon and Meson.

HARDONS:

-> Those particles which experiences strong nuclear force are called hadron.

-> Hadrons are made up of quarks and anti-quarks held by the strong force with the help of field particle gluons.

-> They have integer charge value.

-> example:

proton, neutron, anti-proton, anti-neutron

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- They have no color charge.
- Hadrons are unstable particles (except proton in free space)

They are divided into two Groups

BARYON NUMBER:

A quantum number which is equal to the number of baryons minus the no. of anti-baryons in a system of sub-atomic particles called Baryon Number.

- Baryon Number of quark is 1/3 and anti-quark is -1/3.

MESON:

Those particles having mass between mass of electron and mass of proton are called meson.

- These particles are made up of quark and anti-quark.
- Meson and boson having integral spin (1, 0, -1)
- Their baryon number is "zero."
- All mesons are finally decay into electron, positron, neutrinos and photon.

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Example: Pion ($u\bar{d}$) spin (0) charge (+1)

Kaon ($s\bar{u}$) sth spin (0) charge (-1)

Kaon ($d\bar{s}$) spin (0) charge (0)

BARYON:

Those particles having mass equal to or greater than the mass of proton are called Baryons.

- These particles are made up of three quarks.
- Baryons are fermions having half integer spin (1/2)
- Baryons have Baryon Number +1 and anti-baryons have baryon number -1

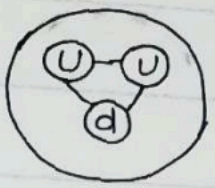
Examples:

Proton (UUd), Neutron (udd), omega (sss), delta (UUU).

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COMPOSITION OF PROTON:

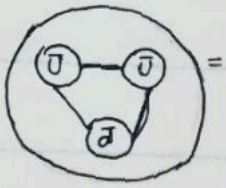
A proton is a baryon consist of 2 up quarks and one down quark. Up quark has charge $+2/3e$ and down quark has charge $+1/3e$.

P ⟶ 

$\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +1e$

A proton in term of color change red up, blue up and green down ~~quarks~~ quarks.

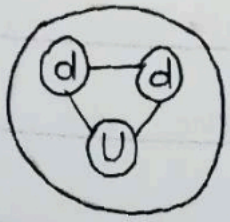
⇒ Anti-proton consist of Anti-quark of same type



= anti-proton

COMPOSITION OF NEUTRON:

A neutron is a baryon made up of 1 up quark and 2-down quark. Up quark has charge $+2/3e$, and down quark has charge $-1/3e$.



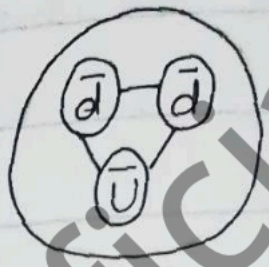
$+\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e = 0$

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A neutron in term of color change below down, red down and a green up quark.

⇒ Anti-neutron consist of antiquark of same type.



Anti-neutron.

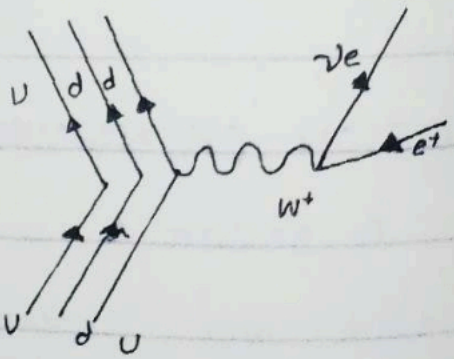
CHANGE IN QUARKS DURING BETA DECAY:

In the Beta positive decay a proton is converted into neutron, positron (β^+) and a neutrino.

$$p = n + e' + \nu e$$

In terms of quarks

$$uud = udd + e^+ + \nu e$$



So in this reaction one up quark turn into a down quark with production of W^+ weak boson and shortly converted into positron and neutrino due to weak nuclear force interaction.

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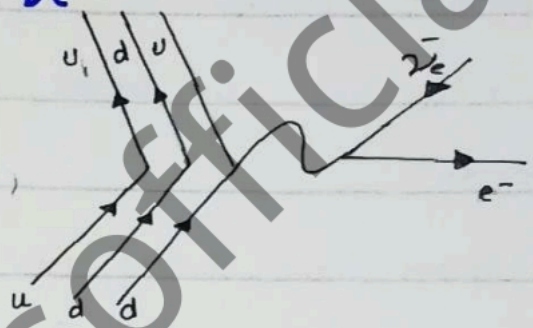
BETA-NEGATIVE DECAY:

In Beta negative decay a neutron is converted into proton, electron and anti-neutrino

$$n \longrightarrow p^+ + e^- + \bar{\nu}_e$$

In term of quark

$$udd = uud + e^- + \bar{\nu}e$$



so in this reaction a down quark is converted into up quark with production of weak boson $\bar{W}$, which shortly converted into electron and anti-neutrino due to weak nuclear force interaction.

LEPTONS:

Leptons are elementary particles which do not experience strong nuclear force.

- leptons are fermions with half integer spin (1/2).
- They are not composed of other particles.
- leptons are not present within the ~~nucleus~~ nucleus, if produced in nucleus, they move out with large K.E.

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-> lepton number for lepton is +1 and for anti-lepton is -1.

-> Second group in standard model of particle physics is lepton.

-> There are six lepton and their six-anti-lepton.

-> Leptons are divided into two main groups

Charged lepton	charge	Neutral lepton	charge
electron e	(-1)	Electron neutrino	0
Muon μ	(-1)	Muon neutrino	0
Tau τ	(-1)	Tau neutrino	0

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PARTICLE PHYSICS

Nuclear Reactions:

The process by which the identity of a nucleus is changed when it is bombarded by an energetic particle is called nuclear reaction.

CONSERVATION LAWS IN NUCLEAR REACTIONS:

1. **Conservation of nucleons:**

The total number of nucleons before and after a nuclear reaction are the same.

2. **Conservation of charge:**

The sum of the charges on all the particles before and after a nuclear reactions are the same.

3. **Conservation of momentum:**

The total momentum of the interacting particles before and after a nuclear reaction is the same.

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TYPES OF NUCLEAR RADIATION

Properties	α-particles	β-particles	γ-rays
Composition	An alpha (α) particle is a high energy helium nucleus. It contains 2 protons and 2 neutrons.	Beta-particles are high-energy electrons (beta-minus) or positron (beta-plus) emitted from radioactive nucleus.	Gamma rays are high-energetic electromagnetic waves emitted fro... exi excited radioacti... nucleus.
Charge	Positively charged particles. Charge on α-particle is +2e.	Negatively charged particles. Charge on each β-particle is e or +e.	Electrically neutral.
Mass	$m_\alpha = 6.64 \times 10^{-27}$ kg	$m_\beta = 9.09 \times 10^{-31}$ kg	zero
Rest mass energy	3.7 GeV	0.511 MeV	zero

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PARTICLE AND ANTI-PARTICLE

British Physicis Paul Dirac predicted, every particle has its own anti-particle. An anti-paiticle has the same mass as the particle version but has opposite charge.

American physicis Carl Anderson observed the positron (which is anti-particle of electron) in a cloud chamber, backing up Dirac's theory.

"Anti-particle have opposite charge, Baryon Number, lepton Number and strangeness."

ELECTRON - POSITRON

Particle:

An electron is a sub-atomic particle with a negative electric charge of -1e. and a mass of approximately 9.11×10^{-31} kg

Anti-particle:

The positron is the antiparticle of the electron with a positive electric charge

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of +le and a mass of approximately 9.11×10^{-31} kg.

Proton- antiproton

Particle:

A proton is a sub-atomic particle found in the nucleus of an atom. It carries a positive electric charge of +le and has a mass of approximately 1.67×10^{-27} kg.

Anti-particle:

The anti-proton is the anti-particle of proton with a negative electric charge of -le and a mass of approximately 1.67×10^{-27} kg. Antiproton can be produced artificially in high energy particle accelerators and are used in scientific experiments to study the fundamental properties of ~~wat~~ matter.

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Alpha, Beta and Gamma decays

ALPHA DECAY:

"A nuclear reaction in which an unstable parent nucleus emits an alpha particle and forms a stable daughter nucleus, is called alpha decay."

EXPLANATION:

When α-particle is emitted out of any nucleus then due to law of conservation of matter, the mass number of the nucleus decreases by 4, and due to law of conservation of charge, the charge of the nucleus decreases by a magnitude of 2e. Alpha decay occurs with nuclei that are very heavy, hence unstable. So, emitting alpha particles, mass number ~~of~~ decreases and it moves towards.

General equation

$$_{Z}X^{A} \longrightarrow {}_{Z-2}Y^{A-4} + {}_{2}He^{4}$$

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Examples:

$$_{92}U^{238} \longrightarrow {}_{90}Th^{234} + {}_{2}He^{4} \quad (\alpha\text{-decay})$$

$$_{88}Ra^{226} \longrightarrow {}_{86}Rn^{222} + {}_{2}He^{4} \quad (\alpha\text{-decay})$$

Beta minus decay:

A neutron in the nucleus is transformed into a proton, an electron and an anti-neutrino.

~~example~~:

Explaination: The anti-neutrino is a nearly massless, electrically neutral particle that carries away some of the energy released during the decay.

=> In beta minus ~~day~~ decay atomic number increases by 1 but mass number remains same.

General equation:

$$_{Z}X^{A} \longrightarrow {}_{Z+1}Y^{A} + \beta^{-} + \bar{\nu}$$

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Example:

$$n \longrightarrow p^{+} + \beta^{-} + \bar{\nu}$$

$${}_{90}Th^{234} \longrightarrow {}_{91}Pa^{234} + {}_{-1}e^{0} + \bar{\nu}$$

$${}_{6}C^{14} \longrightarrow {}_{7}N^{14} + {}_{-1}e^{0} + \bar{\nu}$$

Beta-plus decay:

In beta plus decay, a proton changes into a neutron with the emission of a positron and a neutrino.

Explaination:

The neutrino is a nearly massless, electrically neutral particle that carries away some of the energy released during the decay. In beta plus decay atomic number decreases by 1 but mass number remain same.

General equation:

$${}_{z}X^{A} \longrightarrow {}_{z-1}Y^{A} + \beta^{+} + \nu$$

example: $p^{+} \longrightarrow n + \beta + \nu$

$${}_{7}N^{13} \longrightarrow {}_{6}C^{13} + \beta^{+} + \nu$$

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GAMMA DECAY:

When a nucleus is placed in an excited state, after emission of α or a β decay, it can decay to the ground state by emission of one or more photons called gamma rays and this process is called gamma decay.

General equation:

$$ {}_{z}X^{A} \longrightarrow ({}_{z}X^{A})^{*} + {}_{z}X^{A} + \gamma $$

In gamma decay mass number and atomic number does not change.

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